

Solutions: Let the dot product see

Instructor copy · Exercise session 3

PROBLEM 1

A familiar flight, a different route

Write $c = \vec{u} \cdot \vec{g} = -ug \sin \alpha$. Uniform acceleration gives $\vec{r} = \vec{u}t + \frac{1}{2}\vec{g}t^2$. Take its dot product with itself:

$$S(t) = \vec{r} \cdot \vec{r} = u^2t^2 + ct^3 + \frac{1}{4}g^2t^4, \quad S'(t) = 2u^2t + 3ct^2 + g^2t^3.$$

Height is the projection opposite to gravity: $\vec{g} \cdot \vec{r} = -gh$. The combination $2S - tS'$ eliminates the u^2t^2 term and gives

$$2S - tS' = -ct^3 - \frac{1}{2}g^2t^4 = -t^2\vec{g} \cdot \vec{r} = gt^2h.$$

Thus height can be recovered from squared position and its first derivative:

$$h(t) = \frac{2S - tS'}{gt^2} = -\frac{c}{g}t - \frac{1}{2}gt^2 \quad (t > 0).$$

No horizontal or vertical component equations have been used.

a) Maximum height. Complete the square in the recovered height:

$$h(t) = \frac{c^2}{2g^3} - \frac{g}{2} \left(t + \frac{c}{g^2} \right)^2.$$

Therefore

$$H = \frac{c^2}{2g^3} = \frac{u^2 \sin^2 \alpha}{2g}, \quad t_{\text{top}} = -\frac{c}{g^2} = \frac{u \sin \alpha}{g}.$$

b) Total flight time. Set the recovered height to zero and take the nonzero root:

$$T = -\frac{2c}{g^2} = \frac{2u \sin \alpha}{g} = 2t_{\text{top}}.$$

Optional reflection. Substitute $t_{\text{top}} = -c/g^2$ into S' to obtain

$$S'(t_{\text{top}}) = 2t_{\text{top}}u^2 \cos^2 \alpha > 0.$$

The projectile is still moving horizontally away from the launch point. Maximum height is not maximum distance from the origin. A strictly vertical throw, excluded in the question, would give $S' = 0$ at the top.

Reasoning checkpoint. A squared position contains information about height, but its stationary points answer a different geometric question.

PROBLEM 2

Three speeds, an unseen force

a) Recover scalar information without choosing axes. Let $A = \vec{a} \cdot \vec{a}$ and $B = \vec{v}_0 \cdot \vec{a}$. Taking the dot product of $\vec{v} = \vec{v}_0 + \vec{a}t$ with itself gives

$$v^2 = v_0^2 + 2Bt + At^2.$$

The three readings give, with SI units understood,

$$v_0^2 = 25, \quad 9 = 25 + 2B + A, \quad 25 = 25 + 4B + 4A.$$

Thus $A = 16$ and $B = -16$, so

$$a = 4 \text{ m/s}^2, \quad |\vec{F}| = ma = 8 \text{ N}.$$

If ϕ is the angle between the initial velocity and the force,

$$\cos \phi = \frac{\vec{v}_0 \cdot \vec{a}}{v_0 a} = -\frac{4}{5}, \quad \phi \approx 126.9^\circ.$$

The absolute direction is undetermined: rotating the entire motion across the table leaves every speed reading unchanged.

b) Complete the square. With t in seconds,

$$v^2 = 25 - 32t + 16t^2 = 9 + 16(t - 1)^2.$$

Therefore

$$t_{\min} = 1 \text{ s}, \quad v_{\min} = 3 \text{ m/s}.$$

The puck never stops. At the minimum, $\vec{v} \cdot \vec{a} = B + At = 0$: velocity and acceleration are perpendicular, and neither is zero.

c) Recover a distance without a trajectory. Write $\tau = 1 \text{ s}$. Constant acceleration gives

$$\Delta \vec{r} = 2\tau \vec{v}_0 + \frac{1}{2} \vec{a} (2\tau)^2 = 2\tau (\vec{v}_0 + \vec{a}\tau) = 2\tau \vec{v}(\tau).$$

Taking its squared length gives

$$|\Delta \vec{r}| = 2\tau |\vec{v}(\tau)| = 6 \text{ m}.$$

Since $\vec{v}(\tau) \cdot \vec{a} = 0$, also $\Delta \vec{r} \cdot \vec{F} = 0$: the displacement is perpendicular to the constant force. This is the separation of the endpoints, not the length of the curved path.

Reasoning checkpoint. The readings hide the vector directions, but their squared lengths retain enough information to recover a force magnitude and an angle.

PROBLEM 3

What force turned it back?

Optional · Modeling challenge

a) Let the dot product reveal the force. For radial motion, $\vec{v} = \dot{r} \hat{r}$ and $\vec{F} = F(r) \hat{r}$. Dot Newton's law with velocity:

$$\frac{m}{2} \frac{d(v^2)}{dt} = \vec{F} \cdot \vec{v} = F(r) \dot{r}.$$

On either branch where $\dot{r} \neq 0$, the chain rule gives

$$F(r) = \frac{m}{2} \frac{d(v^2)}{dr}.$$

The positive slope means repulsion, even on approach, when $\dot{r} < 0$. The result extends to the turning point by continuity for a smooth force.

b) Infer the model, then the force. Write $u^2 - v^2 = D/r^p$. The deficits at ℓ , 2ℓ , and 4ℓ are u^2 , $u^2/8$, and $u^2/64$. Doubling separation divides the deficit by eight, so $2^p = 8$, $p = 3$, and $D = u^2 \ell^3$. Hence

$$v^2(r) = u^2 \left(1 - \frac{\ell^3}{r^3} \right), \quad F(r) = \frac{3mu^2 \ell^3}{2r^4} = \frac{C}{r^4}.$$

Here $C = 3mu^2 \ell^3/2$ is inferred from the original motion. Doubling separation divides the force by **sixteen**. The readings determine this model within the assumed power-law family; finitely many readings do not determine an arbitrary function uniquely.

c) Predict a new turning point. Keep C fixed: changing the incoming speed does not change the source. Integrating $d(v^2)/dr = 2C/(mr^4)$ with incoming speed w at infinity gives

$$v^2 = w^2 - \frac{2C}{3mr^3}.$$

At closest approach $v = 0$. With $w = 2u$,

$$r_{\min}^3 = \frac{2C}{3m(2u)^2} = \frac{\ell^3}{4}, \quad r_{\min} = 2^{-2/3} \ell \approx 0.630 \ell.$$

This is an **extrapolation** below the smallest tabulated separation. It assumes the inferred force law continues to hold there. The original expression for v^2 applies only to the original incoming speed.

d) Zero power does not imply zero force. At a turning point $\vec{v} = \vec{0}$, so $d(v^2)/dt = 2\vec{v} \cdot \vec{a} = 0$ even when acceleration is nonzero. For the original approach,

$$F(\ell) = \frac{3mu^2}{2\ell} > 0.$$

The nonzero repulsion reverses the motion. Dividing the power equation by $\dot{r}$ exactly at the turn would be invalid; the force follows from the model and its continuous limit.

Reasoning checkpoint. Motion can reveal a force law. A new prediction tests whether that law extends beyond the data used to infer it.